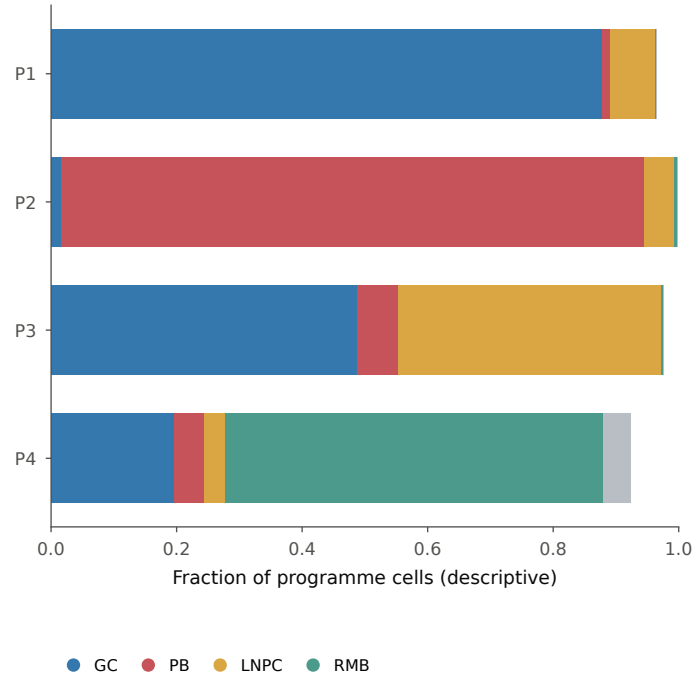
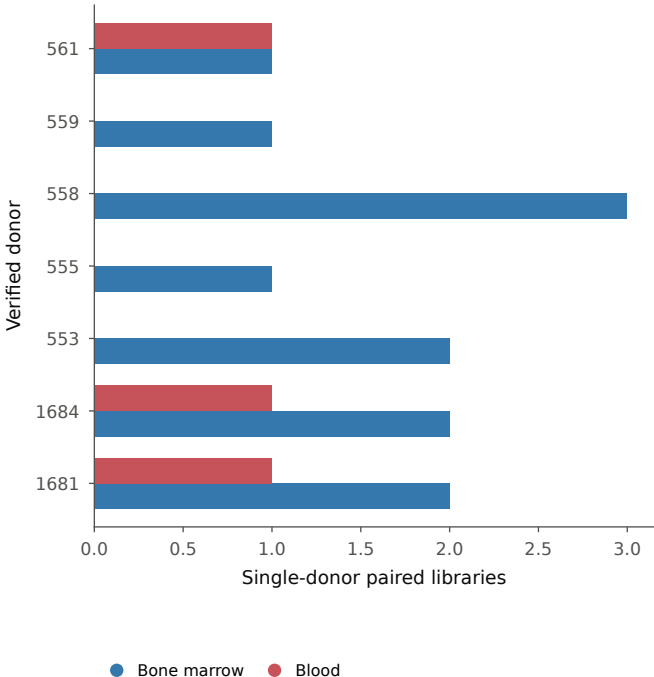


Supplementary Figure 4 | GC and terminal-state biology are complementary records

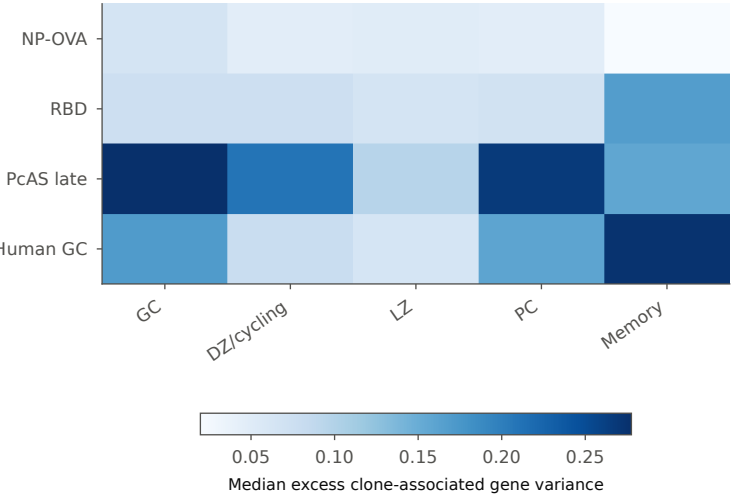
A Longitudinal GC programme composition



B Marrow sampling: explicit donor and sort identity



C Clone-associated expression modules



D Lineage and current state remain distinct

Reporter experiments: 923 tested clones
Within-clone SHM distance did not predict
current expression distance.

Power calibration applies to the two reporter
experiments, not all GC/output datasets.

A clone profile describes a family-wide bias.
A sequence tree describes mutation history.
Neither alone assigns a future cell fate.

E Validation hierarchy

Measured reporter / FACS labels → same-donor sequence membership → conditional controls → external functional test.
Expression signatures and annotation fractions are interpretations of the measured transcriptome, not held-out outcomes.